

## Practice Exercises

1)  $0.\overline{3} = ?$

$$10x = 3.33\ldots$$

$$-x = 0.33\ldots$$

---

$$9x = 3$$

$$x = \frac{3}{9}$$

2)  $0.\overline{142857} = ?$

$$\begin{array}{r} 142857 \\ \hline 999999 \end{array}$$

---

Day - 13

20-01-2026

Find the square, square root  
cube root.

- 1) How to square a number?
- 2) How to find square root?
- 3) How to find cube root?



1) How To square :

$$(88)^2 = \begin{array}{|c|} \hline 8-64 \\ \hline \times 2 \rightarrow 64 \times 2 = 128 \\ \hline 8-64 \\ \hline \end{array}$$

$$\begin{array}{r} 64 \quad 64 \\ 12 \quad 8 \\ \hline 77 \quad 44 \end{array}$$

$(78)^2$

$$\begin{array}{r} 7 \times 8 \\ \swarrow \quad \downarrow \quad \searrow \\ 49 \quad 14 \quad 64 \\ \downarrow \quad \downarrow \\ 11 \quad 2 \\ \hline 60 \quad 84 \end{array}$$

$(63)^2$

$$\begin{array}{r} 6 \times 3 \\ \swarrow \quad \downarrow \quad \searrow \\ 36 \quad 18 \quad 09 \\ \downarrow \quad \downarrow \\ 3 \quad 6 \\ \hline 39 \quad 69 \end{array}$$

$(124)^2$

$$\begin{array}{r} 12 \times 4 \\ \swarrow \quad \downarrow \quad \searrow \\ 144 \quad 24 \quad 16 \\ \downarrow \quad \downarrow \quad \downarrow \\ 12 \quad 4 \quad 16 \\ \hline 153 \quad 76 \end{array}$$

Note

$$\begin{array}{cc} a^2 & b^2 \\ + 2ab & \end{array}$$

$$\begin{array}{r} 13 \\ \wedge \\ 1 \quad 16 \\ 3 \\ \hline 196 \end{array}$$

$$\begin{array}{r} 12 \times 4 \\ \hline 56 \end{array}$$

$(37)^2 =$

$$\begin{array}{r} 37 \\ \swarrow \quad \downarrow \quad \searrow \\ 9 \quad 21 \quad 49 \\ \downarrow \quad \downarrow \\ 42 \\ \hline 1369 \end{array}$$

$(144)^2 =$

$$\begin{array}{r} 144 \\ \swarrow \quad \downarrow \quad \searrow \\ 196 \quad 288 \quad 16 \\ \downarrow \quad \downarrow \\ 11 \quad 2 \\ \hline 20736 \end{array}$$

$$\begin{array}{r} 11 \quad 2 \\ \hline 20736 \end{array}$$

$(154)^2 =$

$$\begin{array}{r} 154 \\ \swarrow \quad \downarrow \quad \searrow \\ 225 \quad 308 \quad 16 \\ \downarrow \quad \downarrow \quad \downarrow \\ 12 \quad 4 \quad 16 \\ \hline 23716 \end{array}$$



How To Find Square Root?

Square

cube

$$1) \sqrt{361} \leftarrow \begin{array}{r} 11 \\ 19 \end{array} \rightarrow 225$$

$$\sqrt{361} = 19$$

$$2) \sqrt{1936}$$

$$19 \overline{) 36} \leftarrow \begin{array}{r} 44 \\ 46 \end{array} \rightarrow 45 = 2025$$

$$\sqrt{1936} = 44$$

$$3) \sqrt{7921} \leftarrow \begin{array}{r} 81 \\ 89 \end{array} \rightarrow 85$$

$$\sqrt{7921} = 89$$

$$4) \sqrt{1764}$$

$$17 \overline{) 64} \leftarrow \begin{array}{r} 42 \\ 48 \end{array} \rightarrow 45 = 2025$$

$$\sqrt{1764} = 42$$

$$1^2 = 1$$

$$2^2 = 4$$

$$3^2 = 9$$

$$4^2 = 16$$

$$5^2 = 25$$

$$6^2 = 36$$

$$7^2 = 49$$

$$8^2 = 64$$

$$9^2 = 81$$

$$10^2 = 100$$

$$1^3 = 1$$

$$2^3 = 8$$

$$3^3 = 27$$

$$4^3 = 64$$

$$5^3 = 125$$

$$6^3 = 216$$

$$7^3 = 343$$

$$8^3 = 512$$

$$9^3 = 729$$

$$10^3 = 1000$$

$$45$$

$$16 \quad 25$$

$$\begin{array}{r} 40 \\ 20 \overline{) 25} \end{array}$$

$$85$$

$$1 \quad 64 \quad 25$$

$$\begin{array}{r} 80 \\ 72 \overline{) 25} \end{array}$$

How To Find Cube Root:

$$1) \sqrt[3]{17576}$$

$$\sqrt[3]{17576} = 26^3$$

$$2) \sqrt[3]{4096} = 16^3$$

$$3) \sqrt[3]{238328} = (62)^3$$

Practice:

$$1) \sqrt{1024} = \begin{matrix} \nearrow 32 \\ \searrow 38 \end{matrix} \rightarrow 35 \Rightarrow 1225$$

$$\begin{array}{r} 35 \\ \swarrow \searrow \\ 9 \quad 25 \\ \hline 30 \\ \hline 1225 \end{array}$$

$$\sqrt{1024} = 32$$

$$2) \sqrt{2025} \Rightarrow 45$$

$$3) \sqrt{681} \begin{matrix} \nearrow 41 \\ \searrow 49 \end{matrix} \rightarrow 45 \Rightarrow 2025$$

$$\sqrt{681} \Rightarrow 41$$

$$4) \sqrt{1521} \begin{matrix} \nearrow 39 \\ \searrow 37 \end{matrix} \rightarrow 35 \Rightarrow 1225$$

$$\boxed{\sqrt{1521} = 39}$$

$$5) \sqrt{1225} \Rightarrow 35$$



Cube roots

$$1) \sqrt[3]{1728} \Rightarrow 12$$

$$2) \sqrt[3]{2197} \Rightarrow 13$$

$$3) \sqrt[3]{2744} = 14$$

$$4) \sqrt[3]{3375} = 15$$

$$5) \sqrt[3]{4096} = 16$$